

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE [SCORE-II]**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	C	B	B	A	D	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	B	C	B	A	D	C	B	A

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	D	A	B	B	B	B	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	A	B	A	D	C	C	A	B

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	A	B	D	A	D	D	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	A	A	B	B	C	D	D	A

PART-1 : PHYSICS**SOLUTION****SECTION-I**

1. Ans. (A)

$$\text{Sol. } \Delta S = nC \ln \frac{T_b}{T} + \frac{mL}{T_b} = 697 \text{ J/K}$$

2. Ans. (C)

$$\text{Sol. } m \frac{dx}{dt^2} + \gamma m \frac{dx}{dt} + kx = F_0 \cos \omega t$$

Now solve

3. Ans. (D)

$$\text{Sol. } f_r = m\omega^2 r$$

$$f_s = m\omega r$$

$$\tan \theta = \frac{f_s}{f_r}$$

4. Ans. (C)

$$\text{Sol. Check all options : } E = \frac{1}{2} nRT$$

5. Ans. (B)

$$\text{Sol. } B = \frac{\mu_0 n i}{2} \left(\cos \theta_1 + \cos \theta_2 \right)$$

$$\& n = \frac{1}{\ell}$$

6. Ans. (B)

Sol. Force between two charges will not change due to presence of 3rd charge.

7. Ans. (A)

$$\text{Sol. } U = -\vec{\mu}_2 \cdot \vec{B}_1 \& B_1 = \frac{\mu_0}{4\pi} \frac{2\mu_1}{r^3}$$

$$F = -\frac{dU}{dr}$$

8. Ans. (D)

$$\text{Sol. } KE = \frac{1}{2} m\omega^2 (A^2 - x^2)$$

$$PE = \frac{1}{2} m\omega^2 x^2$$

$$\therefore KE = \frac{1}{2} m\omega^2 A^2 + PE$$

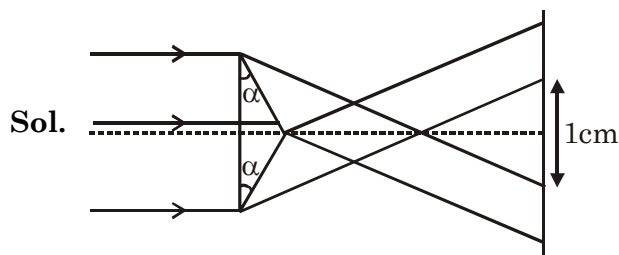
9. Ans. (C)

Sol. step 1 : Find equilibrium position x_0 .

step 2 : Find U at x_0

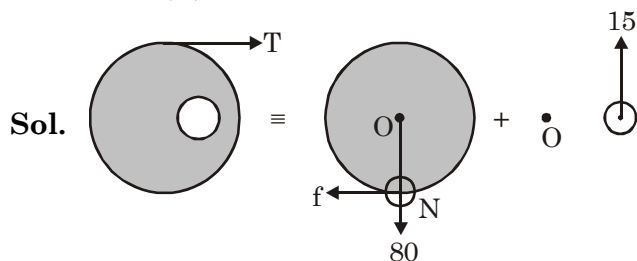
step 3 : work done = $U_f - U_i = 0 - U$

10. Ans. (A)



Now solve

11. Ans. (A)



$$15 \times \frac{2a}{5} = T \cdot 2a \quad \therefore T = 3$$

12. Ans. (A)

13. Ans. (B)

Sol. $mg - T = ma$

$$T = ma \quad \therefore T = mg/2$$

$$a = g/2$$

14. Ans. (C)

Sol. The projectile is initially at semi-minor axis

$\therefore$ Satellite is at equi distance from both focuses.

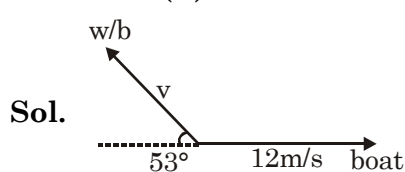
15. Ans. (B)

Sol. Phase diff between top & middle is π .

& middle & bottom is π .

$\therefore$ between top & bottom is 2π .

16. Ans. (A)



$$v \cos 53^\circ = 12 \quad \Rightarrow v \times \frac{3}{5} = 12$$

$$v = 20$$

$$\therefore v \sin 53^\circ = 20 \times \frac{4}{5} = 16 \text{ m/s}$$

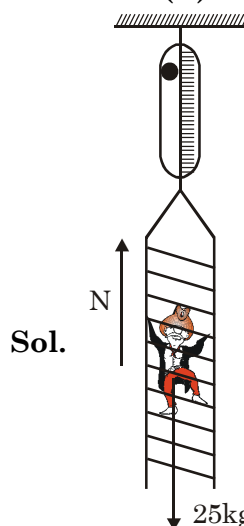
17. Ans. (D)

Sol. If the helicopter is at terminal velocity, then it is falling at constant speed. As such, $h = vT$, where v is the terminal velocity. In that case, $\alpha = 1$.

18. Ans. (C)

Sol. Charge on inner most shell is $-\left(\frac{r_{\text{inner}}}{r_{\text{outer}}}\right)Q$.

19. Ans. (B)



$$N - 25g = 25 \times 1$$

$$N = 275$$

$$\therefore W = 275 + 50 = 325 \text{ N}$$

20. Ans. (A)

Sol. Since the rod will rotate

$\therefore$ velocity will change direction atleast

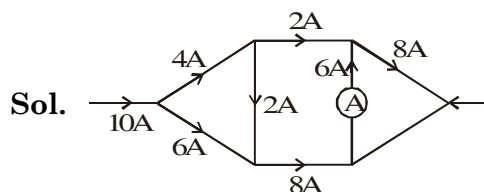
SECTION-II

1. Ans. 0.75

Sol. $E = \frac{\sigma}{4\epsilon_0}$ due to spherical shell.

$$E = \int_R^{2R} \frac{\rho 2\pi r^2 dr}{4\epsilon_0}, \text{ where } \rho = \frac{\rho_0 r}{R}$$

2. Ans. 6.00



3. Ans. 0.05

$$\text{Sol. } M_\theta = \frac{1.5^\circ}{0.25^\circ} = \frac{f_\theta}{f_e} \quad \& \quad f_\theta + f_e = 35$$

4. Ans. 4.00

$$\text{Sol. } T = 2\pi \sqrt{\frac{I}{\mu B}}$$

$$\therefore T \propto \sqrt{I}$$

5. Ans. 0.50

$$\text{Sol. } e = \frac{v_f}{v_i} = \frac{\sqrt{2gh}}{\sqrt{4gh}} = \frac{1}{\sqrt{2}}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans. (D)**

Sol. No. of spherical nodes = $n - \ell - 1$

$$\Rightarrow 3 - 1 - 1 = 1 \text{ for } 3p_x$$

$$\Rightarrow 4 - 2 - 1 = 1 \text{ for } 4d_{x^2-y^2}$$

2. **Ans. (B)**

Sol. At this high $P = 40$ atm

$$V_{\text{gas}} \gg a$$

$$\therefore \frac{a}{V_m^2} \ll P$$

$$\therefore \text{equation becomes } P(V_m - b) = RT$$

$$\therefore T = \frac{P(V_m - b)}{R} = \frac{40(4 - 0.2)}{0.082}$$

$$= 1854 \text{ K}$$

3. **Ans. (B)**

Sol. $K_p = K_c (RT)^{\Delta_g n}$

$$K_p > K_c \text{ if } \Delta_g n > 0 \text{ \& } T > \frac{1}{R}; \left\{ \frac{1}{R} = 12.19 \right\}$$

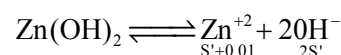
$$K_p > K_c \text{ if } \Delta_g n < 0 \text{ \& } T < \frac{1}{R}$$

4. **Ans. (D)**

Sol. $\text{Zn(OH)}_2 \rightleftharpoons \text{Zn}^{+2} + 2\text{OH}^-$

$$S = (k_{sp}/4)^{1/3} = 10^{-4}$$

In presence of 0.01 M Zn^{+2} .



$$k_{sp} = 0.01 \times (2S)^2$$

$$S = \sqrt{4 \times 10^{-12} / 10^{-2} \times 4} = 10^{-5}$$

5. **Ans. (A)**

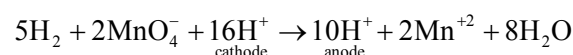
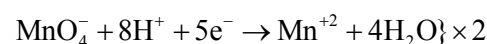
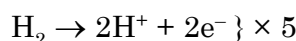
Sol : Fact or process dependent.

6. **Ans. (B)**

Sol. $\text{CMC} \propto \frac{1}{\text{Vander waal force between hydrophobic part of surfactant}}$

7. **Ans. (B)**

Sol. cell R_k^n



$$E_{\text{cell}} = (1.51 - 0) - \frac{0.06}{10} \log \frac{[\text{Mn}^{+2}]^2 [\text{H}^+]_{\text{anode}}^{10}}{[\text{MnO}_4^-]^2 \cdot [\text{H}^+]_{\text{cathode}}^{16} (P_{\text{H}_2})^5}$$

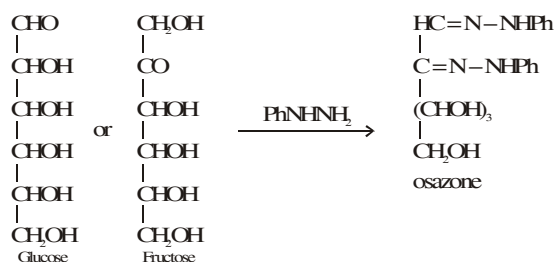
$$= 1.51 - \frac{0.06}{10} \log \frac{(10^{-1})^2 (10^{-1})^{10}}{(0.1)^2 (1)^{16} \times 1}$$

$$= 1.51 - \frac{0.06}{10} \log \frac{10^{-12}}{10^{-2}} = 1.51 + \frac{0.06}{10} \times 10$$

$$= 1.57 \text{ V}$$

8. **Ans. (B)**

Sol. D(+)-glucose and D(-)-fructose forms same osazone on reaction with PhNHNH_2

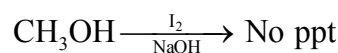
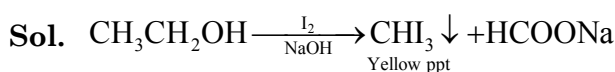


9. **Ans. (B)**

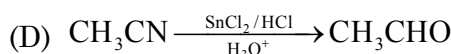
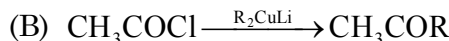
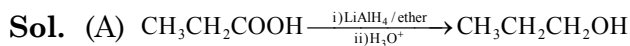
Sol. $\text{CH}_3\text{COOCH}_2\text{CH}_3$



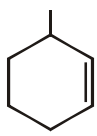
10. **Ans. (C)**



11. **Ans. (D)**



12. **Ans. (C)**

Sol. Compound  has non super imposable mirror image.

13. **Ans. (A)**

Sol. NBS carryout allylic substitution of H atom by free radical mechanism.

ALLEN

14. Ans. (B)

Sol. CN^- is strong field ligand due to which Fe^{+3} configuration will be t_{2g}^5 hence 1 unpaired electron whereas in $[\text{FeF}_6]^{3-}$, F^- is weak field ligand hence due to weak field splitting, configuration will be $t_{2g}^3 e_g^2$ i.e. 5 unpaired electrons.

15. Ans. (A)

Sol. The intermediate formed in reaction of tetrahalides with H_2O is sp^3d hybridised and carbon does NOT have vacant d-orbitals.

16. Ans. (D)

Sol. H_2O : triatomic and hence planar

BrF_3 : Bent T-shaped

XeF_4 : Square planar

17. Ans. (C)

Sol. $\text{Li}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{LiOH}$,

$\text{KO}_2 + \text{H}_2\text{O} \rightarrow \text{KOH} + \text{H}_2\text{O}_2 + \text{O}_2$

$\text{Na}_2\text{O}_2 + \text{H}_2\text{O} \rightarrow \text{NaOH} + \text{H}_2\text{O}_2$

18. Ans. (C)

Sol. ZnS - White

PbS - Black

HgO - Yellow

19. Ans. (A)

Sol. Cinnabar- HgS , Calamine- ZnCO_3

Carnallite- $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

Zincite- ZnO

20. Ans. (B)

Sol. $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{PH}_3$

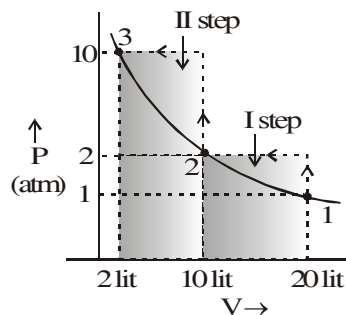
$\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2 + \text{NH}_3$

$\text{CaC}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{C}_2\text{H}_2$

SECTION-II

1. Ans. 100.00

Sol.



If vol. is 10 lit

$$P_2 = \frac{1 \times 20}{10} = 2 \text{ atm}$$

If $V = 2$ lit

$$\text{then } P_3 = \frac{2 \times 10}{2} = 10 \text{ atm}$$

$$W = (20-10) \times 2 + (10-2) \times 10 = 100 \text{ lit} \cdot \text{atm}$$

2. Ans. 2.65

$$\text{Sol. } \alpha = \sqrt{\frac{k_a}{c}} = \sqrt{\frac{6.4 \times 10^{-5}}{10^{-2}}}$$

$$= \sqrt{6.4 \times 10^{-3}} = 8 \times 10^{-2} = 0.08$$

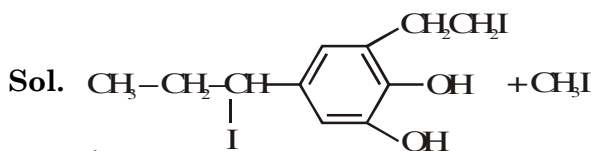
$$\pi = MRT(1 + 0.08) = 0.01 \times 0.082 \times 300 \times 1.08$$

$$= 0.265 \text{ atm}$$

$$\text{Hence } x = 0.265$$

$$x \times 10 = 2.65$$

3. Ans. 3.00



4. Ans. 6.00

5. Ans. 1.00

Sol. Iron has +1 oxidation state in brown ring complex $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

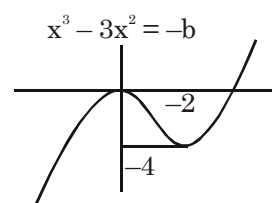
1. **Ans. (B)**

Sol. $y^2 = 4(x+1)$ $x^2 = 4(y+1)$
 $y = m(x+1) + \frac{1}{m}$ $y+1 = mx - m^2$
 $\Rightarrow m = 1$
 $\Rightarrow$ equation of common tangent is
 $x + y + 2 = 0$

Area = 2

2. **Ans. (D)**

Sol. $f(x) = x^3 + ax^2 + b$
 $f'(x) = 3x^2 + 2ax$ at $x=2$ is 0
 $12 + 4a = 0$
 $a = -3$



$-4 < -b < 0$
 $\Rightarrow 0 < b < 4$

3. **Ans. (C)**

Sol. Equation of normal at any point

$P\left(ct, \frac{c}{t}\right)$ on $xy = c^2$ is $xt^3 - yt - ct^4 + c = 0$

If it passes through $Q(h, k)$ then

$ct^4 - ht^3 + kt - c = 0$

$\Sigma x_1 = \Sigma ct_1 = h, \Sigma y_1 = \Sigma \frac{c}{t} = k, x_1x_2x_3x_4 = -c^4$

$\frac{\Sigma x_1 \Sigma y_1}{x_1x_2x_3x_4} = -\frac{hk}{c^4}$

If $h = 3, k = 2, c = 2$

$-\frac{hk}{c^4} = -\frac{6}{16} = -\frac{3}{8}$

4. **Ans. (A)**

Sol. Tangent

$\frac{x \cos \theta}{12} + \frac{y \sin \theta}{9} = 1$

$A(12 \sec \theta, 0) \quad B(0, 9 \operatorname{cosec} \theta)$

$m = AB = \sqrt{144 \sec^2 \theta + 81 \operatorname{cosec}^2 \theta}$

$= \sqrt{225 + (12 \tan \theta - 9 \cot \theta)^2 + 216}$

$= 21$

Normal

$\frac{12x}{\cos \theta} - \frac{9y}{\sin \theta} = 63$

$P(0,0) = M = \frac{63}{\sqrt{144 \sec^2 \theta + 81 \operatorname{cosec}^2 \theta}}$

$Mm = 63$

5. **Ans. (B)**

Sol. $\beta + \gamma = \frac{\pi}{2}$

$\cos \beta = \sin \gamma$

$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$\cos^2 \alpha = 0$

$(\cos \alpha + \sin \beta + \sin \gamma)^2$

$= \sin^2 \beta + \sin^2 \gamma + 2 \sin \beta \sin \gamma$

$= 1 + \sin 2\beta$

$= 2 \text{ max value.}$

6. **Ans. (D)**

Sol. expand determinant

7. **Ans. (A)**

Sol. $x = 1 \quad y = -1 \quad z = 1$

8. **Ans. (D)**

Sol. $x^2 + \frac{1}{x^2} = 2 \cos \theta$

$\cos \theta = 1 \quad x^2 = 1$

$\theta = 0, 2\pi \quad x = \pm 1$

9. **Ans. (D)**

Sol. $\frac{\Sigma x_i^2}{n} \geq \left(\frac{\Sigma x_i}{n}\right)^2 \Rightarrow \frac{200}{n} \geq \frac{(40)^2}{n^2} \Rightarrow n > 8$

10. **Ans. (A)**

Sol. Mode = 3 median - 2 mean

$= 3 \times 26 - 2 \times 25 = 78 - 50 = 28$

11. **Ans. (B)**

Sol. $p \quad q \quad p \leftrightarrow q \quad \sim p \quad \sim p \leftrightarrow q$

$T \quad T \quad T \quad F \quad F$

$T \quad F \quad F \quad F \quad T$

$F \quad T \quad F \quad T \quad T$

$F \quad F \quad T \quad T \quad F$

12. **Ans. (A)**

Sol. by fundamental principal of counting 2^{n^2-n} .

13. **Ans. (A)**

Sol. $U_{10} = \int_0^{\pi/2} x^{10} \sin x dx$

$= \left[-x^{10} \cos x \right]_0^{\pi/2} + \int_0^{\pi/2} 10x^9 \cos x dx$

$$U_{10} = 10 \left(x^9 \sin x \right)_0^{\pi/2} - 90 \int_0^{\pi/2} x^8 \sin x dx$$

$$U_{10} = 10 \left(\frac{\pi}{2} \right)^9 - 90 U_8$$

$$U_{10} + 90 U_8 = 10 \left(\frac{\pi}{2} \right)^9$$

14. Ans. (A)

$$\text{Sol. } \int_0^1 f(x) dx + \int_0^1 f(1+x) dx + \dots + \int_0^1 f(7+x) dx$$

$$\int_0^1 f(x) dx + \int_1^2 f(x) dx + \dots + \int_7^8 f(x) dx$$

$$= \int_0^8 f(x) dx = 3$$

15. Ans. (B)

$$\text{Sol. } \int_0^1 [f(x) - f(2x)] dx = 5 \quad \dots (i)$$

$$\int_0^2 [f(x) - f(4x)] dx = 10 \quad \dots (ii)$$

replace x by 2t in (ii)

$$x = 2t$$

$$dx = 2dt$$

$$2 \int_0^1 [f(2t) - f(8t)] dt = 10 \quad \dots (iii)$$

2 × equation (i) + equation (iii)

$$2 \int_0^1 [f(x) - f(8x)] dx = 20$$

16. Ans. (B)

$$\text{Sol. } \Rightarrow \int_0^a f(a-x)g(a-x)h(a-x) dx$$

$$= k \int_0^a f(x)g(x) dx$$

$$\Rightarrow \int_0^a f(x)g(x) \cdot \left(\frac{3h(x)-5}{4} \right) dx = k \int_0^a f(x)g(x) dx$$

$$\Rightarrow \frac{3}{4} \int_0^a f(x)g(x)h(x) dx - \frac{5}{4} \int_0^a f(x)g(x) dx$$

$$= k \int_0^a f(x)g(x) dx$$

$$\Rightarrow \frac{3}{4} \left[k \int_0^a f(x)g(x) dx \right] - \frac{5}{4} \int_0^a f(x)g(x) dx$$

$$= k \int_0^a f(x)g(x) dx$$

$$\Rightarrow \left(\frac{3}{4}k - \frac{5}{4} - k \right) \int_0^a f(x)g(x) dx = 0$$

$$\frac{3}{4}k - \frac{5}{4} - k = 0$$

$$k = -5$$

17. Ans. (C)

$$\text{Sol. } \alpha = \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3}$$

$$\beta = \frac{1}{n^4} \left[\left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6} \right]$$

$$= \frac{n^2(n+1)^2}{4n^4} - \frac{n(n+1)(2n+1)}{6n^4}$$

$$\beta = \frac{1}{4}$$

$$\Rightarrow 4\beta = 3\alpha$$

18. Ans. (D)

$$\text{Sol. } \lim_{x \rightarrow 0} f(x) = a^3 \text{ independent of } x$$

19. Ans. (D)

$$\text{Sol. } \lim_{x \rightarrow \frac{1}{\sqrt{2}}} f(x) - 2 \text{ must be zero for finite limit.}$$

$$\therefore \lim_{x \rightarrow \frac{1}{\sqrt{2}}} f(x) - 2 = 0$$

$$\lim_{x \rightarrow \frac{1}{\sqrt{2}}} f(x) = 2$$

$$\therefore \lim_{x \rightarrow \frac{1}{\sqrt{2}}} e^{f(x)} = e^2$$

20. Ans. (A)

$$\text{Sol. } \left(\frac{3x}{5} \right)^2 + \left(\frac{4y}{5} \right)^2 = 1$$

$$9x^2 + 16y^2 = 25$$

Point of intersection with $y = x^2$ is (1,1)

$$\left(\frac{dy}{dx} \right) = 2x = 2 = m_1 \left(\frac{dy}{dx} \right)_{\text{at}(1,1)} = \frac{-9}{16} = m_2$$

$$\tan \theta = \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right) = \left(\frac{2 + \frac{9}{16}}{1 - \frac{9}{16} \cdot 2} \right) = \frac{41}{2}$$

$$Q = \tan^{-1} \left(\frac{41}{2} \right)$$

SECTION-II

1. Ans. 7.00

$$\text{Sol. } \left. \begin{array}{l} a - 4b + c = 0 \\ \& ap + bq + c = 0 \end{array} \right\} \Rightarrow p = 1, q = -4$$

2. Ans. 42.00

Sol. $[\log 1 + \log 4 + \log 9 + \dots + \log 100] - 2\log 7!$

$$= 2\log(1 \cdot 2 \cdot 3 \cdot \dots \cdot 10) - 2\log 7!$$

$$= 2\left[\log \frac{10!}{7!}\right] = 2\log 720 = 2\log(10 \times 9 \times 8)$$

$$= 2[1 + 3\log 2 + 2\log 3]$$

$$= 2(1 + 3x + 2y) = 2 + 6x + 4y$$

$$\therefore a = 2, b = 6, c = 4$$

$$2a + 3b + 5c = 4 + 18 + 20 = 42$$

3. Ans. 2.00

Sol. $\vec{s} \cdot (\vec{p} + \vec{q} + \vec{r}) = 16$

$$(x + y + z)[\vec{p} \ \vec{q} \ \vec{r}] = 16$$

$$x + y + z = 2$$

4. Ans. 12.00

Sol. Only possibility for max

$$\begin{pmatrix} 2 & & & & \\ & 2 & & & \\ & & 2 & & \\ & & & 2 & \\ & & & & 2 \end{pmatrix}$$

5. Ans. 48.00

Sol. $f'(3) = 6$

$$\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = 6$$

$$\lim_{h \rightarrow 0} \frac{f(3+h)}{h} = 6$$

$$\Rightarrow \lim_{3x \rightarrow 0} \frac{f(3+3x)}{3x} \times 3 + \lim_{5x \rightarrow 0} \frac{f(3+5x)}{5x} \times 5$$

$$6 \times 3 + 6 \times 5 = 48$$